Phase Transitions and Synchronization in the Kuramoto Model

Term Paper 2 – IDC621 Modelling Complex Systems

Siddhartha Rao

Abstract

This report investigates the synchronization dynamics and phase transitions of coupled oscillators using the Kuramoto model. By employing a mean-field approach and a fourth-order Runge-Kutta (RK4) integration scheme, we simulate the time evolution of the system's complex order parameter. The study explores synchronization thresholds across standard normal, uniform, and Lorentzian natural frequency distributions, validating empirical results against theoretical critical coupling values (K_c). Furthermore, we analyze the distinction between phase-locked and drifting oscillators to characterize the macroscopic steady-state behavior of the complex system.

1 Introduction

The Kuramoto model is a foundational mathematical paradigm for studying synchronization in complex systems, describing phenomena ranging from the flashing of fireflies to the firing of pacemaker neurons and the synchronization of power grids. It represents a large population of coupled limit-cycle oscillators, where each oscillator has its own intrinsic natural frequency but is weakly coupled to all others.

The standard governing equation for N oscillators is given by:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i), \quad i = 1, \ldots, N \tag{1}$$

where θ_i is the phase of the i-th oscillator, ω_i is its natural frequency, and $K \geq 0$ is the global coupling strength. As K increases, the system undergoes a phase transition from an incoherent state to a partially or fully synchronized state.

2 Mathematical Framework and Methodology

2.1 Mean-Field Reduction

To measure the macroscopic synchronization of the system, we define the complex order parameter, $re^{i\psi}$, which represents the centroid of the phases on a unit circle:

$$re^{i\psi} = \frac{1}{N}\sum_{j=1}^{N} e^{i\theta_j} \tag{2}$$

Here, $r(t) \in [0, 1]$ measures the phase coherence, and $\psi(t)$ is the average phase. Multiplying both sides by $e^{-i\theta_i}$ and taking the imaginary part allows us to rewrite the governing equation in the mean-field form:

$$\frac{d\theta_i}{dt} = \omega_i + Kr\sin(\psi - \theta_i) \tag{3}$$

This formulation is computationally advantageous, reducing the numerical complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$.

2.2 Numerical Integration

The system was simulated using a custom Python implementation. To ensure high numerical stability and accuracy over thousands of time steps, a fourth-order Runge-Kutta (RK4) method was utilized with a time step of $dt = 0.05$ or $dt = 0.1$, depending on the specific experimental bounds.

3 Results and Discussion

3.1 System Dynamics and Phase Synchronization

The time evolution of the system ($N = 500$) was observed under varying coupling strengths: subcritical ($K = 0.5$), transitional ($K = 1.5$), and supercritical ($K = 3.0$).

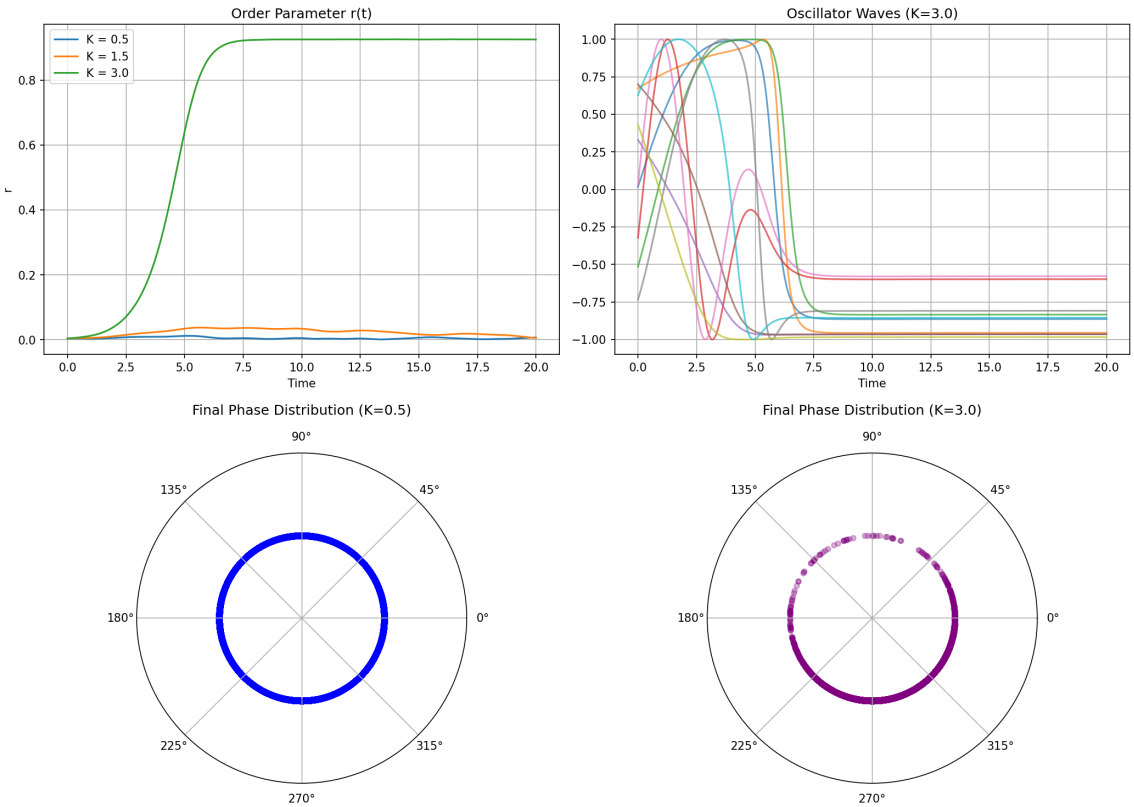

Figure 1: System dynamics. Top left: Time evolution of the order parameter $r(t)$. Top right: Phase trajectories of 10 selected oscillators at $K = 3.0$. Bottom: Final phase distributions on the unit circle for weak ($K = 0.5$) and strong ($K = 3.0$) coupling.

As shown in Figure 1, for $K = 0.5$, the order parameter remains near zero, and the final phases are uniformly distributed, indicating incoherence. For $K = 3.0$, $r(t)$ rapidly approaches 1, and the phases collapse into a single synchronized cluster.

3.2 Phase Transitions Across Frequency Distributions

The critical coupling strength K_c at which the onset of synchronization occurs depends heavily on the probability density function $g(\omega)$ of the natural frequencies. According to Kuramoto's continuous limit theory, $K_c = 2/(\pi g(0))$.

We simulated $N = 800$ oscillators across three distributions: Normal, Uniform, and Lorentzian.

- **Normal Distribution** ($\mu = 0, \sigma = 1$): $K_c = \sqrt{8/\pi} \approx 1.59$

- **Uniform Distribution** ($\omega \in [-\sqrt{3}, \sqrt{3}]$): $K_c = 4\sqrt{3}/\pi \approx 2.20$
- **Lorentzian Distribution** ($\gamma = 1.0$): $K_c = 2\gamma = 2.00$

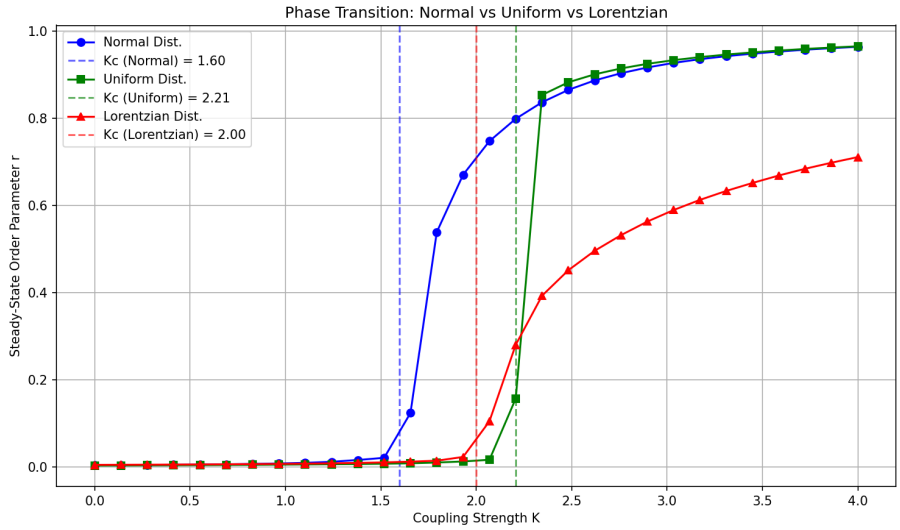

Figure 2: Steady-state order parameter r as a function of coupling strength K for Normal, Uniform, and Lorentzian natural frequency distributions. Dashed lines indicate theoretical K_c values.

Figure 2 demonstrates excellent agreement between our numerical simulations and the analytical predictions for the onset of synchronization. The phase transition is clearly second-order (continuous) for all tested distributions.

3.3 Effective Frequency and Oscillator Locking

In a partially synchronized state, the oscillator population splits into two groups: those that are phase-locked to the mean field, and those that drift. The effective frequency is defined as $\omega_{eff} = \lim_{T\to\infty} \frac{\theta_i(T)-\theta_i(0)}{T}$.

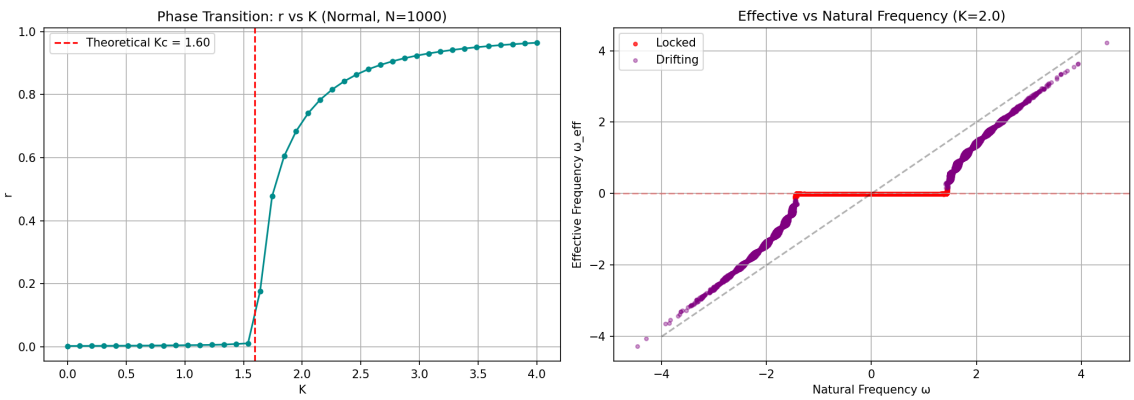

Figure 3: Left: Phase transition curve for $N = 1000$ normally distributed oscillators. Right: Effective frequency ω_{eff} versus natural frequency ω_i at $K = 2.0$, illustrating the locked (red) and drifting (purple) populations.

Figure 3 shows that oscillators with natural frequencies close to the mean frequency ($|\omega_i| < Kr$) are entrained and share a common effective frequency ($\omega_{eff} \approx 0$). Oscillators with larger natural frequencies fail to synchronize and drift, pulling away from the mean field.

4 Conclusion

The numerical simulations of the Kuramoto model successfully capture the complex macroscopic behavior of coupled non-linear oscillators. Using a computationally efficient mean-field reduction and an RK4 integrator, we verified the continuous phase transitions from incoherence to synchronization. The empirical data strongly matched the theoretical critical coupling constants K_c for normal, uniform, and Lorentzian distributions. Furthermore, the bifurcation between locked and drifting oscillator populations was distinctly visualized, highlighting the rich dynamic interplay of complex systems.

Code Availability: The complete Python code and simulation data used to generate these results are available on GitHub: https://github.com/Sidluca28/Kuramoto-Model